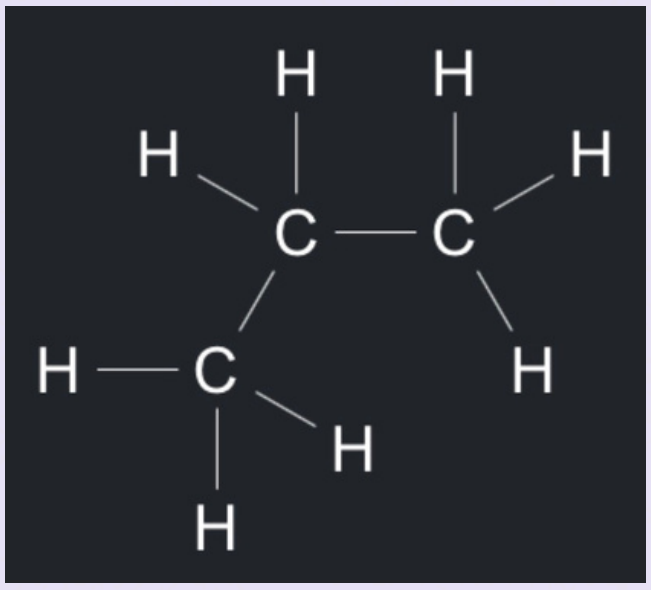
CARD 01 — PROPANE

MOLECULAR FORMULA

C_3H_8

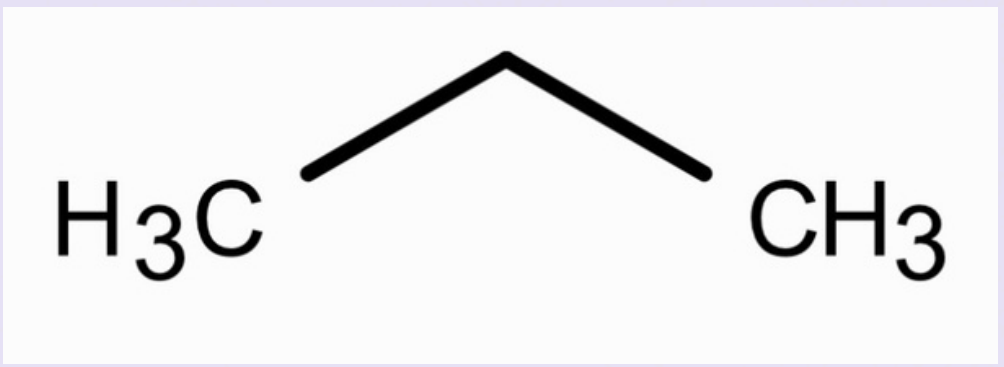
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$CH_3CH_2CH_3$

BOND LINE FORMULA



VERIFICATION STRIP

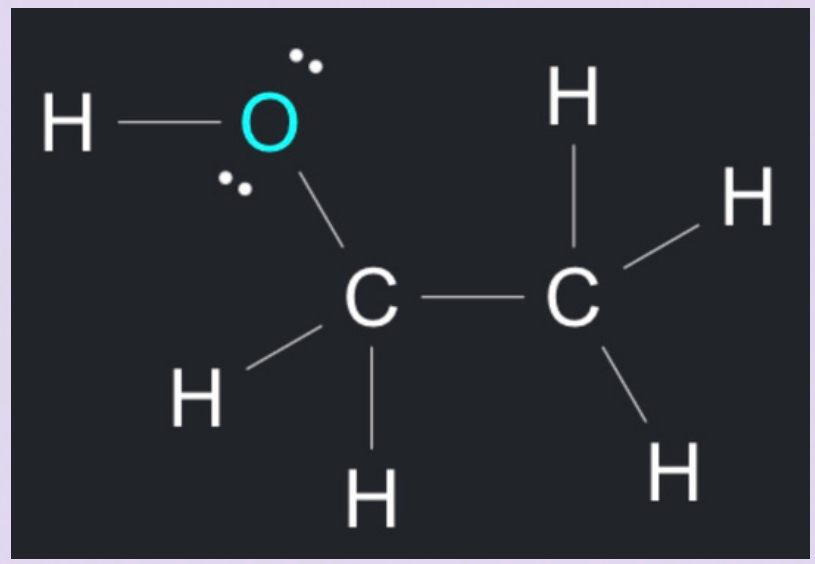
- [x] atom count verified: 3C + 8H
- [x] carbon tetravalency checked
- [x] bond orders and heteroatom valences checked
- [x] same connectivity across all four representations

CARD 02 — ETHANOL

MOLECULAR FORMULA

C_2H_6O

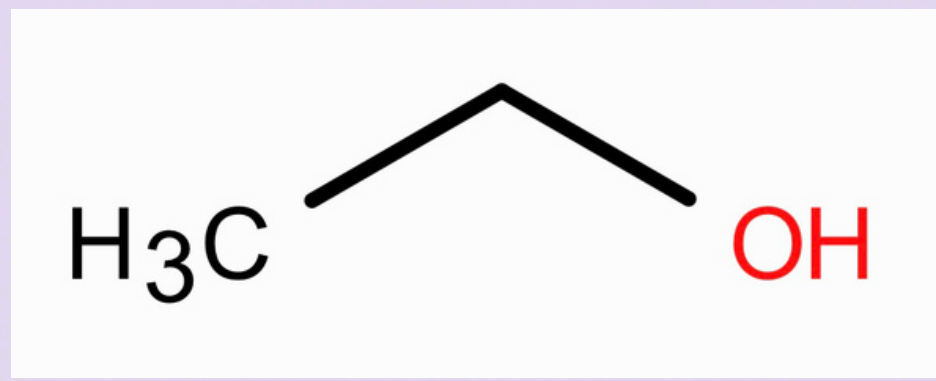
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

CH_3CH_2OH

BOND LINE FORMULA



VERIFICATION STRIP

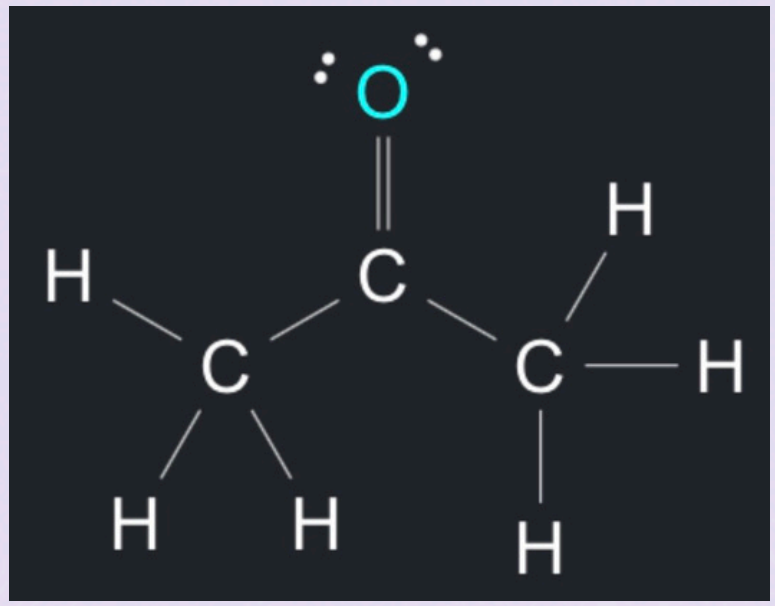
- [x] atom count verified: 2C + 6H + 1O
- [x] carbon tetravalency checked
- [x] bond orders and heteroatom valences checked
- [x] same connectivity across all four representations

CARD 03 — ACETONE

MOLECULAR FORMULA

C_3H_6O

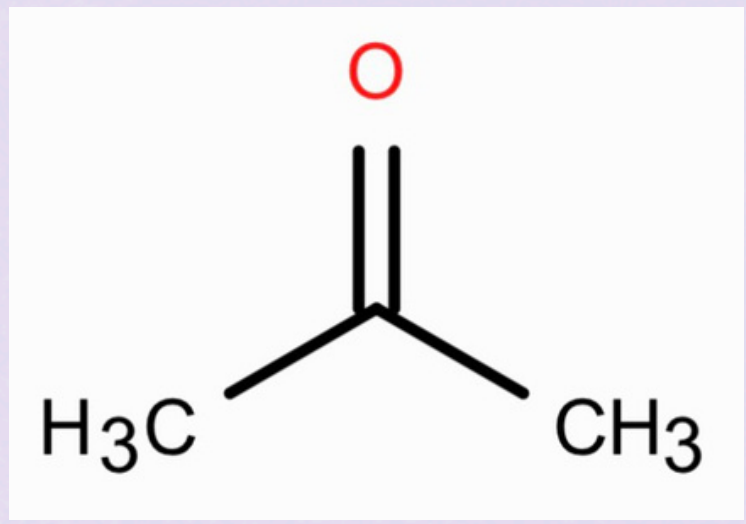
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

CH_3COCH_3

BOND LINE FORMULA



VERIFICATION STRIP

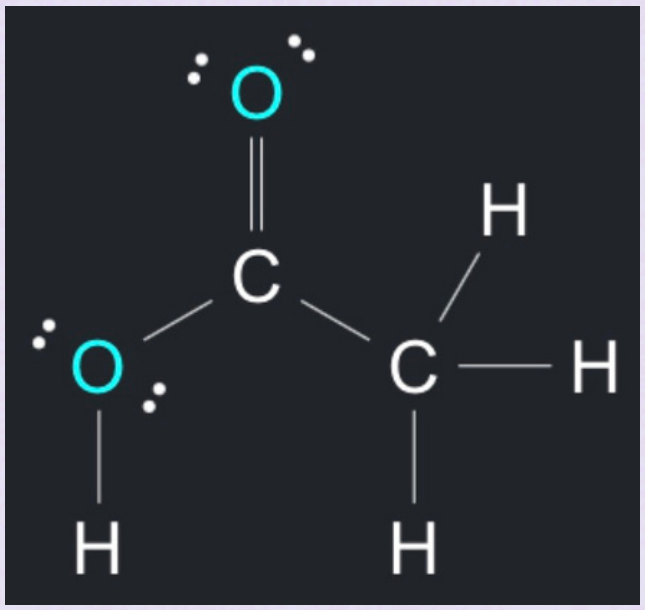
- [x] atom count verified: 3C + 6H + 1O
- [x] carbon tetravalency checked
- [x] bond orders and heteroatom valences checked
- [x] same connectivity across all four representations

CARD 04 — ACETIC ACID

MOLECULAR FORMULA

$$C_2H_4O_2$$

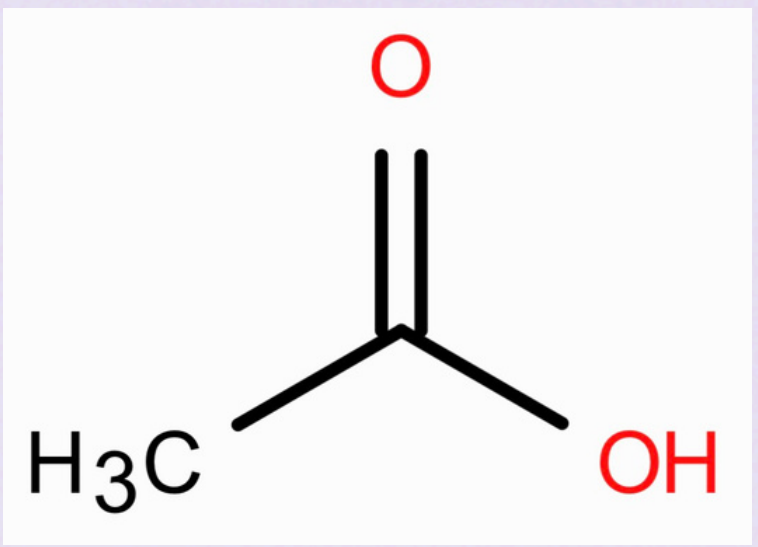
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$CH_3COOH$$

BOND LINE FORMULA



VERIFICATION STRIP

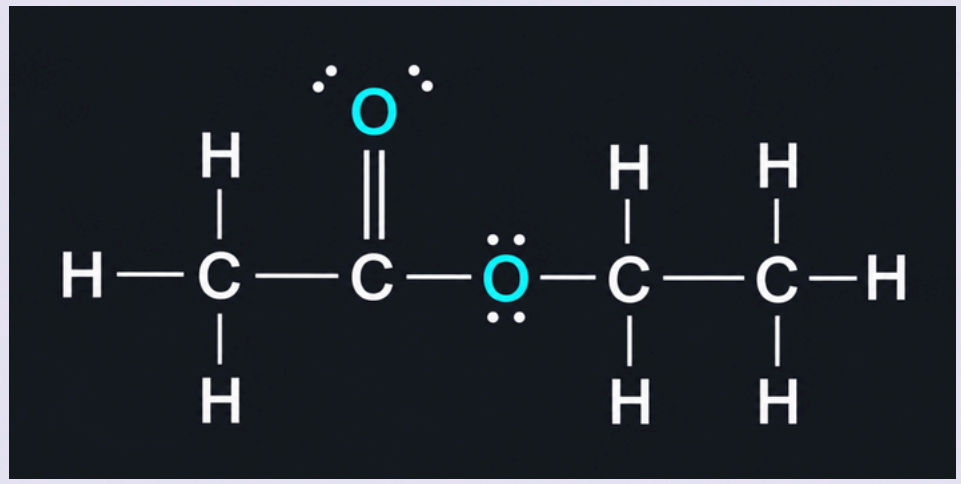
- [x] atom count verified: 2C + 4C +2O
- [x] carbon tetravalency checked
- [x] bond orders and heteroatom valences checked
- [x] same connectivity across all four representations

CARD 05 — ETHYL ACETATE

MOLECULAR FORMULA

$$C_4H_8O_2$$

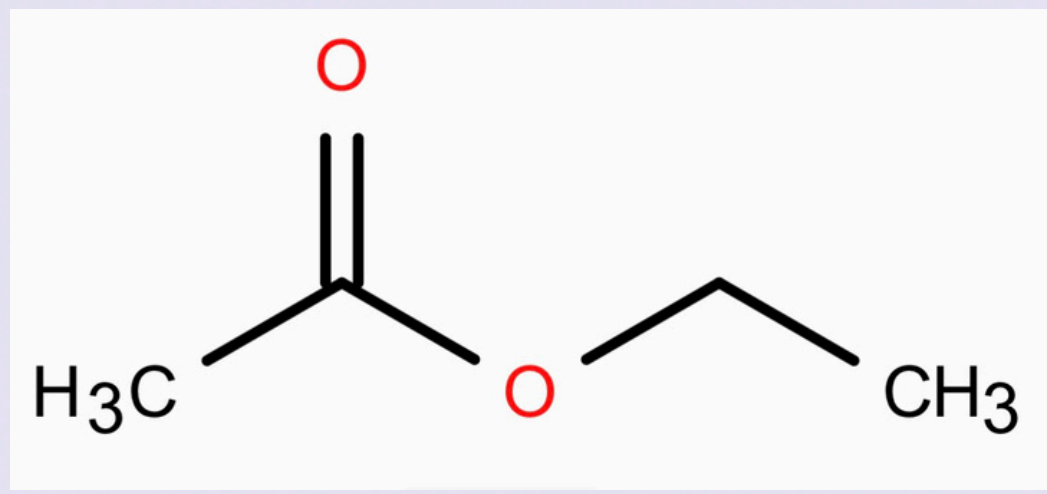
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$CH_3COOCH_2CH_3$$

BOND LINE FORMULA



VERIFICATION STRIP

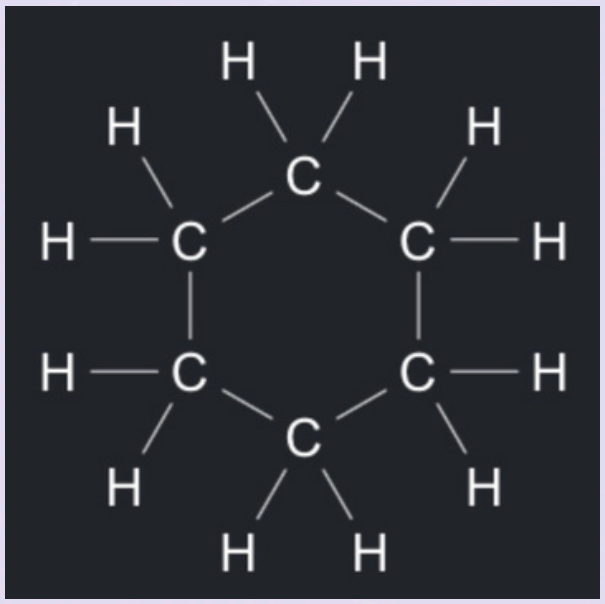
- [x] atom count verified: 4C + 8H + 2O
- [x] carbon tetravalency checked
- [x] bond orders and heteroatom valences checked
- [x] same connectivity across all four representations

CARD 06 — CYCLOHEXANE

MOLECULAR FORMULA

$$C_6H_{12}$$

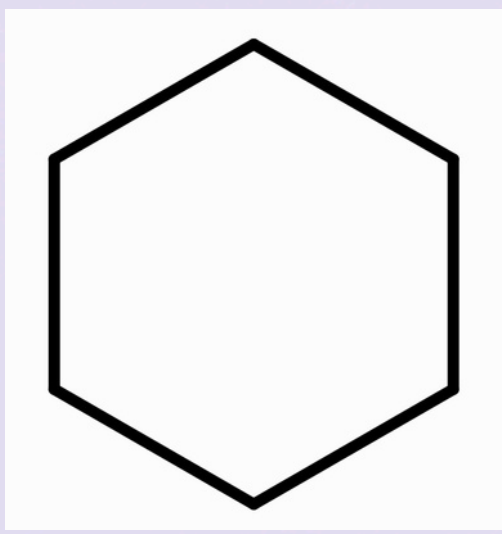
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$(CH_2)_6$$

BOND LINE FORMULA



VERIFICATION STRIP

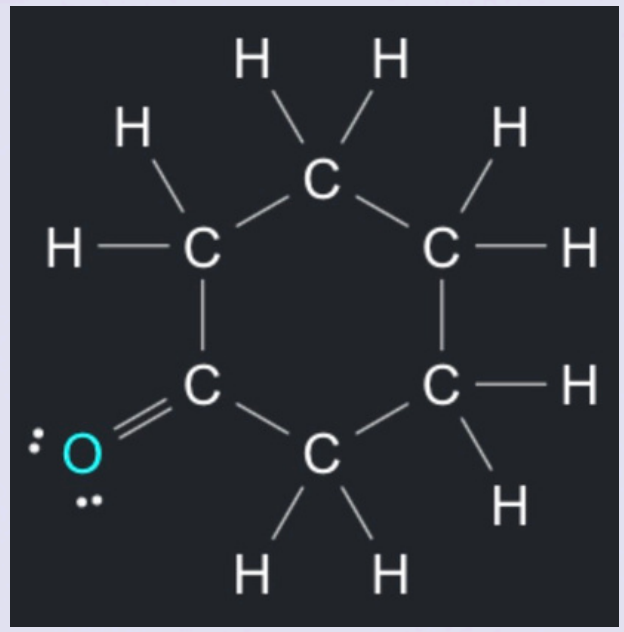
- [x] atom count verified: 6C + 12H
- [x] carbon tetravalency checked
- [x] bond orders and heteroatom valences checked
- [x] same connectivity across all four representations

CARD 07 — CYCLOHEXANONE

MOLECULAR FORMULA

$C_6H_{10}O$

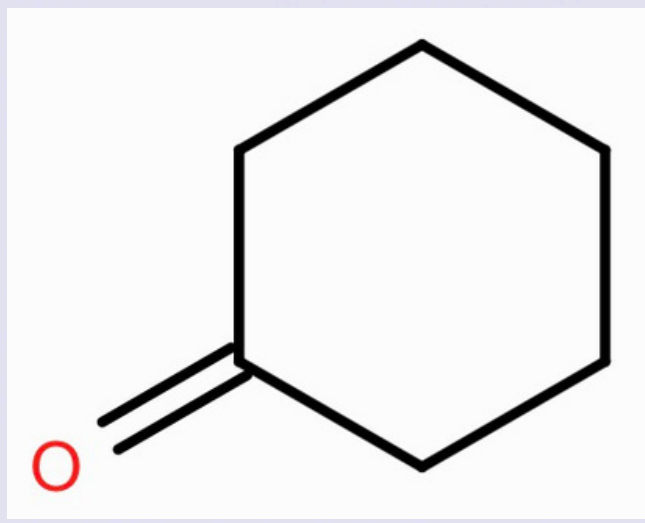
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$(CH_2)_5CO$

BOND LINE FORMULA



VERIFICATION STRIP

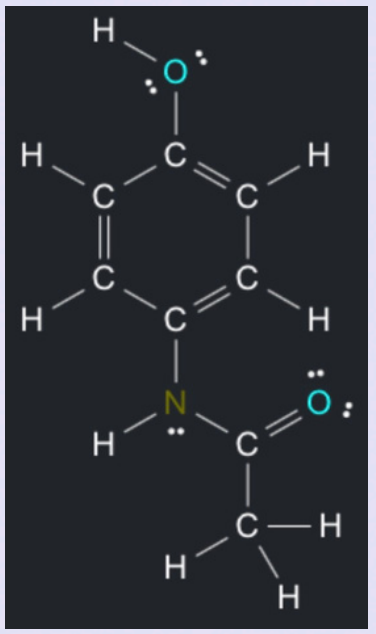
- [x] atom count verified: 6C + 10H + 1O
- [x] carbon tetravalency checked
- [x] bond orders and heteroatom valences checked
- [x] same connectivity across all four representations

CARD 08 — ACETAMINOPHEN

MOLECULAR FORMULA

$$C_8H_9NO_2$$

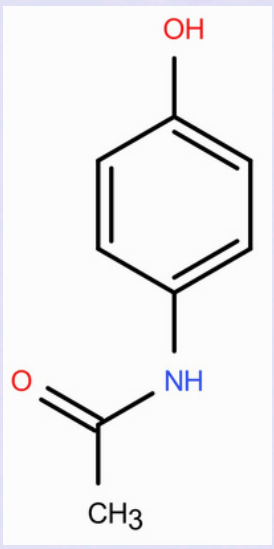
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$HOC_6H_4NHCOCH_3$$

BOND LINE FORMULA



VERIFICATION STRIP

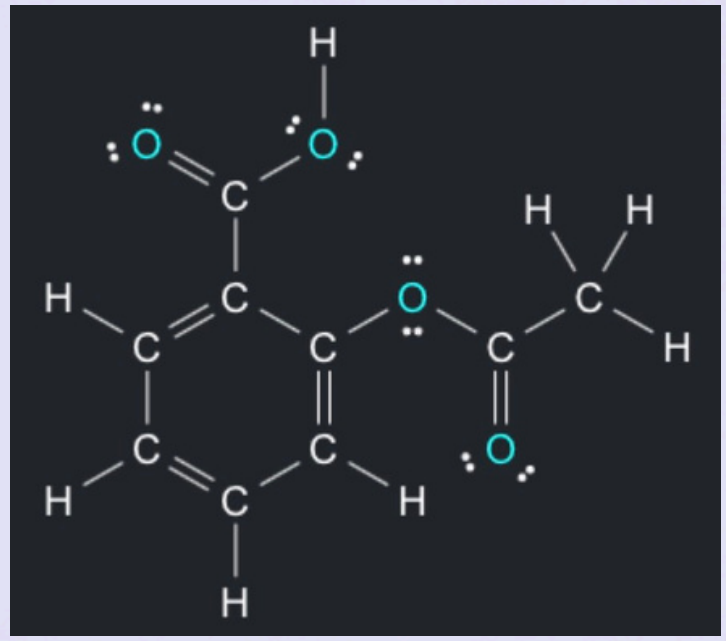
- ☑ atom count verified: 8C + 9H + 1N + 2O
- ☑ carbon tetravalency checked
- ☑ bond orders and heteroatom valences checked
- ☑ same connectivity across all four representations

CARD 09 — ASPIRIN

MOLECULAR FORMULA

$C_9H_8O_4$

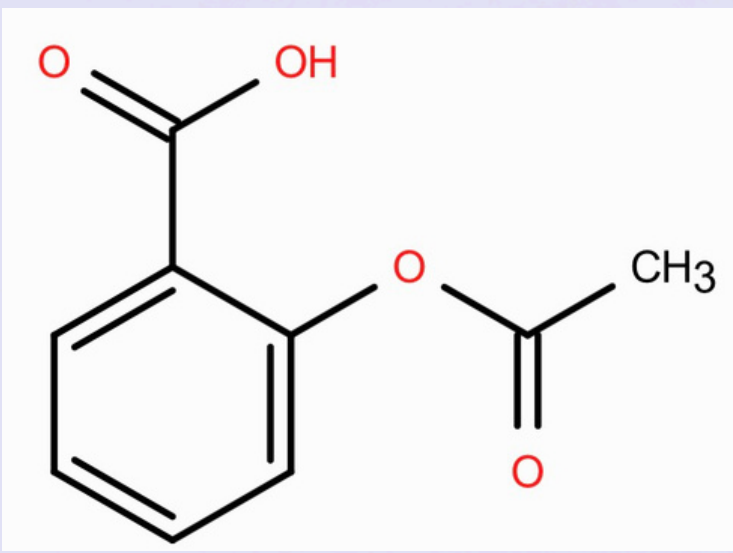
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$CH_3COOC_6H_4COOH$

BOND LINE FORMULA



VERIFICATION STRIP

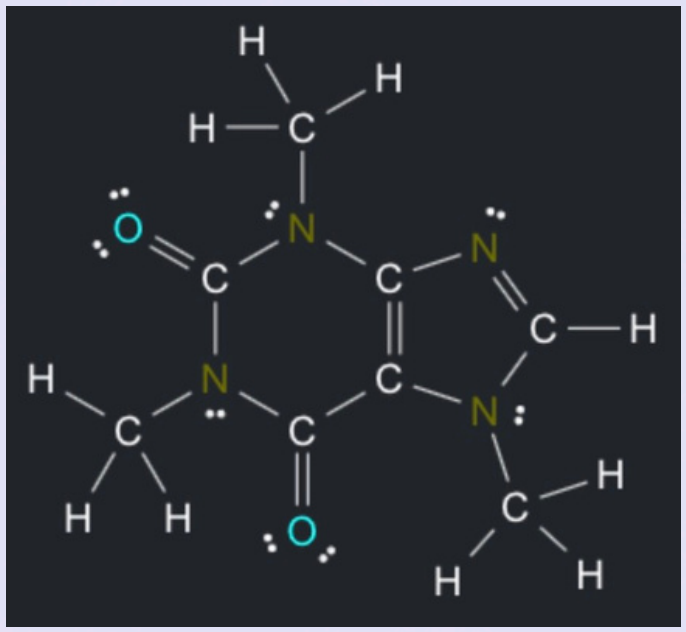
- [x] atom count verified: 9C + 8H + 4O
- [x] carbon tetravalency checked
- [x] bond orders and heteroatom valences checked
- [x] same connectivity across all four representations

CARD 10 — CAFFEINE

MOLECULAR FORMULA

$$C_8H_{10}N_4O_2$$

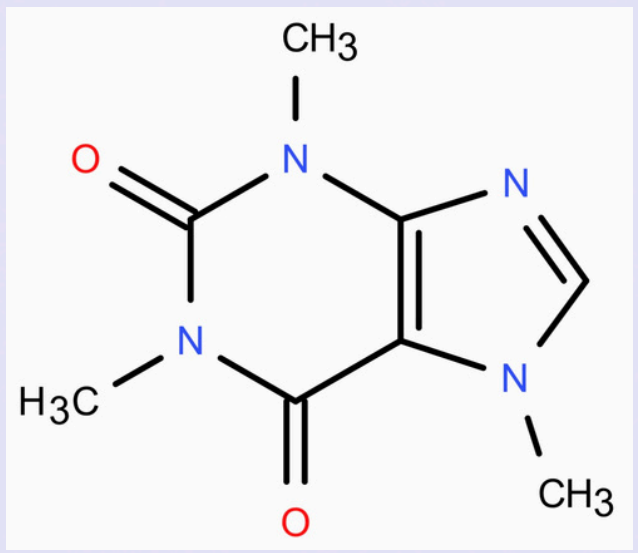
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$(CH_3)_3C_5HN_4O_2$$

BOND LINE FORMULA



VERIFICATION STRIP

- ☑ atom count verified: 8C + 10H + 4N + 2O
- ☑ carbon tetravalency checked
- ☑ bond orders and heteroatom valences checked
- ☑ same connectivity across all four representations